MILITARY SPECIFICATION

RESISTORS, FIXED, FILM
ESTABLISHED RELIABILITY
STYLE RNR55[1/]

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RNR55[1/] established reliability, film, fixed resistors.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification. The following specification, forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of this document shall be that listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-R-55182 - Resistors, Fixed, Film, Established Reliability, General Specification for.

(Copies of the specification, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-55182, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

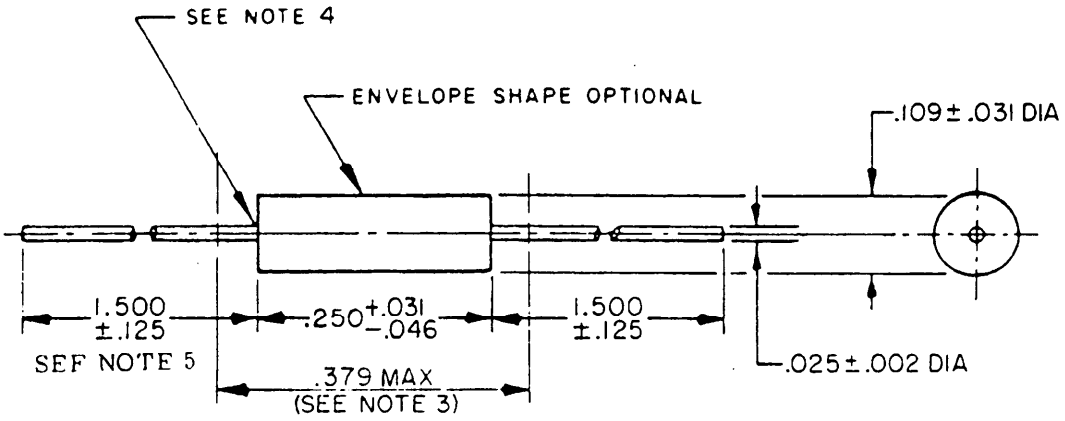
3.2.1 Characteristic. Style RNR55 is available in characteristics C, E, H, J, and K.

3.2.2 Terminal. Terminal types RNR, RNC, and RNN are available in all characteristics. Terminal RNR is inactive for design with characteristics H, J, and K.

1/ Third letter is variable, dependent upon lead material or capability.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, NJ 07703 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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INCHES	MM
.002	0.05
.025	0.64
.031	0.79
.046	1.17
.109	2.77
.125	3.18
.250	6.35
.379	9.63
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Maximum length is clean lead to clean lead.
4. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter.
* 5. Lead length for new design and tape and reel packaging shall be 1.00 + .625, -0.000 inch (25.4 + 15.88, -0.00 mm).

FIGURE 1. Style RNR55 resistors. 1/

1/ See footnote 1, page 1.

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3.3 Power rating. The power rating shall be 0.125 watt (see 6.2).

3.4 Voltage rating. The continuous working voltage shall not exceed 200 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as follows:

Minimum resistance:

Characteristic:

C, E, H, J, K - 10.0 ohms.

Maximum resistance:

Characteristic

H, J, K - 2.00 megohms.

C, E- 1.21 megohms.

3.6 Voltage coefficient (applicable to resistors of 1,000 ohms and above). The voltage coefficient shall not exceed ±0.005 percent per volt.

* 3.7 Insulation. Insulation material surrounding the resistor body shall extend around the entire shoulder of each end cap toward the axis of the lead attachments. Complete back wall coverage is optional.

*3.8 Marking. Due to size limitations this style resistor shall be marked with the following minimum information:

042K - Date code, characteristic.
49R9 - Coded resistance value.
*FMCJ - Tolerance, failure rate, lead material, JAN marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection. Sampling and inspection shall be in accordance with MIL-R-55182.

* 4.2 Qualification conformance inspection. The quality conformance inspection required by MIL-R-55182, verify compliance with 3.7 on a sample size equivalent to the sample required for a visual inspection of group A, subgroup 2. AQL and LQ shall be defined for a major defect for this requirement.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-55182.

5.2 Marking. Packaging shall be marked with assigned part number, manufacturers identification name, logo or code number (FSCM).

6. NOTES

6.1 Notes. The notes specified in MIL-R-55182 are applicable to this specification.

6.2 Power rating at 70°C. The power rating at 70°C for 2,000 hours life test has been established at 0.125 watt, maximum voltage 200 volts. It should be noted that the failure rate level is established at the single condition of 125°C, 0.10 watt and 10,000 hour life test duration.

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DWV-Insulating Sleeves-Method 301 of ... 10 percent.

6.3 Supersession data. This specification supersedes MIL-R-10509/9(USAF), and MIL-R-55182/8(USAF) in the following manner.

Characteristic in MIL-R-55182/1 in FR level "M"	Supersedes characteristic in MIL-R-10509/9(USAF)	Supersedes characteristic in MIL-R-55182/8(USAF)
K	D	---
C or E	G (up to .200 megohm) incl	---
H	C	H
J	E	J

6.4 MIL-R-10509 substitution data. Resistors of this specification, regardless of their failure rate designation, are substitutes for resistors of the same resistance value and tolerance specified in the inactivated specification MIL-R-10509/7. Substitute characteristics and terminal types are indicated in table I.

* 6.5 MIL-R-55182 substitution data. Resistors of this specification, regardless of their failure rate designation, are substitutes for resistors of the same resistance value and tolerance specified in the inactive specification MIL-R-55182/12.

TABLE I. Characteristic and terminal substitution.

MIL-R-10509 Style RN55	MIL-R-55182/1 Substitute characteristic	MIL-R-55182/1 Substitute terminal
Inactive characteristic		
C	H	RNC
E	J	RNC
D	K	RNC
G	C (up to 1.21 megohms)	RNC, RNR

6.6 Equivalent NATO sytles. The equivalent NATO styles per NATO-NEPR NO. 6 for RNR 55 shall be as follows:

Military type 1/	NATO style
RNR 55E	NRN 42
RNR 55H	NRN 21
RNR 55J	NRN 32
RNR 55K	NRN 52

1/ Third letter is variable dependent upon lead material or capability.

6.7 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement NATO-NEPR No. 6. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AR
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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